CANDIDATE NAME.: _____

NANJIRIRI CLUSTER EXAMINATIONS

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATIONS

BIOLOGY

Subject Number: M022/II

WEDNESDAY, 18TH MAY

Time Allowed: 2 hour sessions

8:00 onwards

PAPER II

Practical (40 marks)

Instructions

1. **This paper contains 6 printed pages. Please check.**
2. This paper contains **two** sections, **A** and **B.**
3. Answer **all** the **four** questions in the spaces provided. The maximum number of marks for each answer is indicated against each question. A pencil should be used for all drawings.
4. Write your **Name and School Name** at the top of each page of your question paper in the spaces provided.
5. Do not spend more than **30 minutes** on a single question.
6. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
	Total		

©2022 NANJIRIRI MOCK **Turn over…..**

NAME:_____

2022 **M022/I**

1. Describe an experiment that could be carried out to show that heat is produced during respiration. In an essay form, your answer should include procedure, expected results and conclusion.

(10 marks)

2. Describe an experiment that can be carried out to show that green leaves contain three pigments. In an essay form; your answer should include; procedure, expected results and conclusion.

_____

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(10 marks)

3. You are provided with the following;

- A part of stem marked **X**.
- A part of stem marked **Y**.
- A razor blade.
- 30 cm ruler.

a) To which group of flowering plants does each of the specimen **X** and **Y** belong?

X _____

(1 mark)

Y _____

(1 mark)

b. Draw a cross section of the specimen **X** and label any two parts.

(4 marks)

c). Which type of root system is found in specimen **X** and **Y**?

_____

_____

(2 marks)

d). State any two structural difference between X and Y.

_____

_____ (2 marks)

4. You are provided with the following;

- Irish potato tuber
- Solution in a container labelled **A**, **B** and **C**.
- Razor blade

Procedures

i. Cut eight equal sized strips of Irish potato each measuring 20mm long, 5mm wide and 5mm high.

ii. Put two strips into each of the containers labelled **A**, **B** and **C**.

iii. Leave for 10 minutes

a) After 10 minutes, measure and record the length of the strips in the table below.

Solution	Length of strips (mm)	Average length of the strips (mm)
A **B** **C**		

(6 marks)

b) (i) In which container did the strips decrease in length most

_____

(1 mark)

(ii) Explain your answer to 1b(i) above

_____

(2 marks)

c) Which solution had highest water solution.

_____

(1 mark)

END OF QUESTION PAPER

N.B: This paper contains 6 printed pages.